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pitwall-intelligence/

ROOT

├── README.md   project overview, setup, BVI formula
├── docker-compose.yml   docker   postgres + redis + backend + frontend
├── Makefile   build, test, seed, clean targets
├── .env.example   DB_URL, REDIS_URL, CORS_ORIGINS
└── .gitignore   venv, .env, __pycache__, node_modules

BACKEND (FASTAPI)

├── backend/
│   ├── Dockerfile   python:3.11-slim, uvicorn
│   ├── requirements.txt   py
│   └── app/
│       ├── main.py   py   FastAPI app, middleware, router mount
│       ├── api/   /v1 route handlers
│       │   ├── bvi.py   py   GET /bvi/scores, /bvi/compare
│       │   ├── constructors.py   py   GET /constructors/rankings
│       │   ├── drivers.py   py   GET /drivers/analytics/{id}
│       │   ├── seasons.py   py   GET /season/comparison
│       │   ├── circuits.py   py   GET /circuits/{id}/performance
│       │   ├── predict.py   py   POST /predict/podium
│       │   └── admin.py   py   POST /admin/retrain
│       ├── core/   config, db session, cache
│       │   ├── config.py   py   settings via pydantic-settings
│       │   ├── database.py   py   SQLAlchemy engine + session
│       │   └── cache.py   py   Redis client, 24hr TTL helpers
│       ├── models/   SQLAlchemy ORM + Pydantic schemas
│       │   ├── orm.py   py   Race, Driver, Constructor, Result…
│       │   └── schemas.py   py   Pydantic request/response models
│       ├── services/   BVI engine, ML inference
│       │   ├── bvi_engine.py   py   perf/cons/expo scoring + composite
│       │   ├── ml_registry.py   py   joblib model load/serve singleton
│       │   └── sponsorship.py   py   tier classification logic
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ML & NOTEBOOKS

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├── notebooks/
│   ├── 01_eda.ipynb   exploratory data analysis
│   ├── 02_bvi_computation.ipynb   BVI scoring walkthrough
│   ├── 03_constructor_points_model.ipynb   GBR training + RMSE eval
│   └── 04_podium_classifier.ipynb   logistic regression + AUC-ROC
├── ml_models/   serialised joblib artefacts (.gitignore large files)
│   ├── constructor_points_gbr.joblib
│   └── podium_classifier_lr.joblib
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DATA & ETL

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├── data/
│   ├── raw/   Kaggle CSVs (circuits, drivers, races…)
│   └── processed/   bvi_scores.csv, feature matrices
├── etl/
│   ├── load.py   py   CSV → PostgreSQL via psycopg2 COPY
│   ├── transform.py   py   dedup, normalise points systems
│   └── schema.sql   sql   CREATE TABLE statements, FK constraints
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FRONTEND (REACT)

```
├── frontend/
│   ├── Dockerfile   node build → nginx serve
│   ├── package.json   js   react 18, recharts, tailwind, react-router
│   ├── tailwind.config.js   js
│   ├── nginx.conf   SPA routing fallback
│   └── src/
│       ├── main.jsx   js   React root, BrowserRouter
│       ├── App.jsx   js   layout, nav, route definitions
│       ├── pages/   one file per dashboard tab
│       │   ├── BviOverview.jsx   js
│       │   ├── ConstructorRankings.jsx   js
│       │   ├── DriverAnalytics.jsx   js
│       │   ├── SeasonComparison.jsx   js
│       │   ├── CircuitIntelligence.jsx   js
│       │   ├── PodiumPredictor.jsx   js
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│   │   └── SponsorshipTiers.jsx   js
│   ├── components/   shared UI components
│   │   ├── BviRadarChart.jsx   js
│   │   ├── TierBadge.jsx   js
│   │   ├── ScoreBar.jsx   js
│   │   └── Navbar.jsx   js
│   └── api/
│       └── client.js   js   axios/fetch wrappers per endpoint
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TESTS

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└── tests/
    ├── test_bvi_engine.py   py   unit tests — scoring formula
    ├── test_api_bvi.py   py   integration — /api/v1/bvi/scores
    └── test_predict.py   py   integration — /api/v1/predict/podium
```